

NAME: _____ PDG: _____ Register No: _____



ANDERSON JUNIOR COLLEGE

Preliminary Examination 2010

CHEMISTRY

Higher 1

Paper 2

8872/02

17 Sep 2010

2 hours

Candidates answer Section A on the Question Paper

Additional Materials: Writing paper

Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, PDG and register number on all the work you hand in.

Write in dark blue or black pen.

You may use a pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **ALL** the questions.

Section B

Answer **TWO** questions on separate writing paper.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use						
Paper 1 [30m]						Total [110]
Paper 2 Section A [40m]	1		Paper 2 Section B [40m]	5		
	2			6		
	3			7		
	4			Grade		

This document consists of **16** printed pages including cover page.

Section A

Answer **all** the questions in the space provided.

- 1 (a) Draw a detailed diagram to represent the arrangement of protons, neutrons and electrons in an atom of carbon.

[1]

- (b) Carbon forms double bonds with each of the Group VI elements oxygen, sulfur and selenium. In each case, the double bond is polar. However, the polarities of these double bonds do not necessarily cancel in the molecules that carbon formed with the elements.

	Overall polarity of molecule
CO ₂	0
COS	0.71
COSe	0.73

- (i) With the aid of an appropriate diagram, explain why carbon dioxide molecule has no net polarity.

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- (ii) Why is the COSe molecule more polar than the COS molecule?

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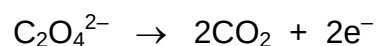
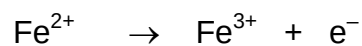
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[4]

(c) Ethanedioate ion ($\text{C}_2\text{O}_4^{2-}$), which is formed on dimerisation of the CO_2^- anion, can react with many metal ions to produce salts. One example is iron(II) ethanedioate (FeC_2O_4) which is used as a yellow dye.

(i) When an aqueous solution of iron(II) ethanedioate reacts with potassium manganate(VII) under acidic condition, $\text{Fe}^{2+}(\text{aq})$ and $\text{C}_2\text{O}_4^{2-}(\text{aq})$ undergo oxidation as shown.



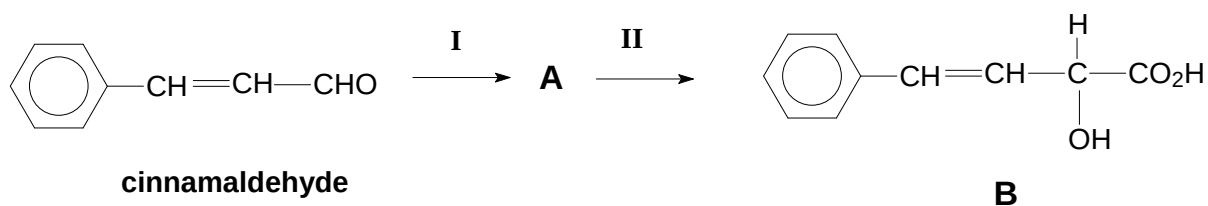
Construct a half-equation for the oxidation of iron(II) ethanedioate under acidic condition.

(ii) Hence, calculate the volume of 0.10 mol dm^{-3} potassium manganate(VII) required to oxidise 25 cm^3 of an acidified solution of 0.10 mol dm^{-3} iron(II) ethanedioate.

[3]

[Total: 8]

- 2 (a) Cinnamaldehyde is the main component in cinnamon oil. It can be converted to acid **B** via intermediate **A** as shown in the reaction pathway below.



- (i) Draw the structural formula for intermediate **A**.
- (ii) State the reagents and conditions for stages **I** and **II** and name the type of reaction that takes place at each stage.

Stage	Reagents and Conditions	Type of reaction
I		
II		

[5]

- (b) Cinnamaldehyde can be reduced into 3-phenylpropan-1-ol. Suggest the reagent and condition for this reaction.

[1]

(c) Anhydrous phosphorus pentachloride, PCl_5 , is an off-white crystalline solid which is often used to identify the presence of the -OH functional group in alcohol.

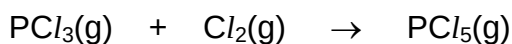
(i) With the aid of an equation, explain how anhydrous PCl_5 may be used to identify the -OH functional group in 3-phenylpropan-1-ol formed in (b).

(You may represent 3-phenylpropan-1-ol by R-OH .)

.....

 [1]

(ii) PCl_5 is produced when PCl_3 reacts with Cl_2 gas. In the *gaseous* phase, the reaction can be represented as follows:

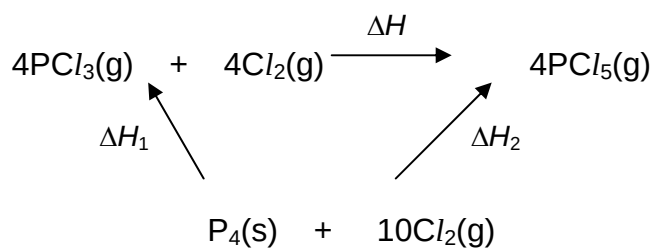


Use the following average bond energies to calculate the enthalpy change of this reaction.

Bond	Average bond energy / kJ mol^{-1}
$\text{Cl} - \text{Cl}$	244
$\text{P} - \text{Cl}$	320

[2]

- (iii) The enthalpy change of the reaction in (c)(ii) can be determined indirectly using the energy cycle shown.



What does ΔH_1 represent?

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What is the relationship between ΔH , ΔH_1 and ΔH_2 ?

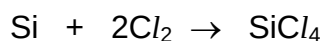
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[2]

[Total:11]

- 3 Chlorine is very reactive and will form compounds by direct combination with many elements or molecules.

- (a) Chlorine reacts with magnesium and silicon to form magnesium chloride and silicon tetrachloride respectively.



The chlorides each dissolves in or react with water to form solutions with different pH values.

- (i) Suggest the approximate pH values of the solution formed in **each** case.

MgCl_2 :

SiCl_4 :

- (ii) Explain the difference between the two pH values, giving relevant equations where appropriate.

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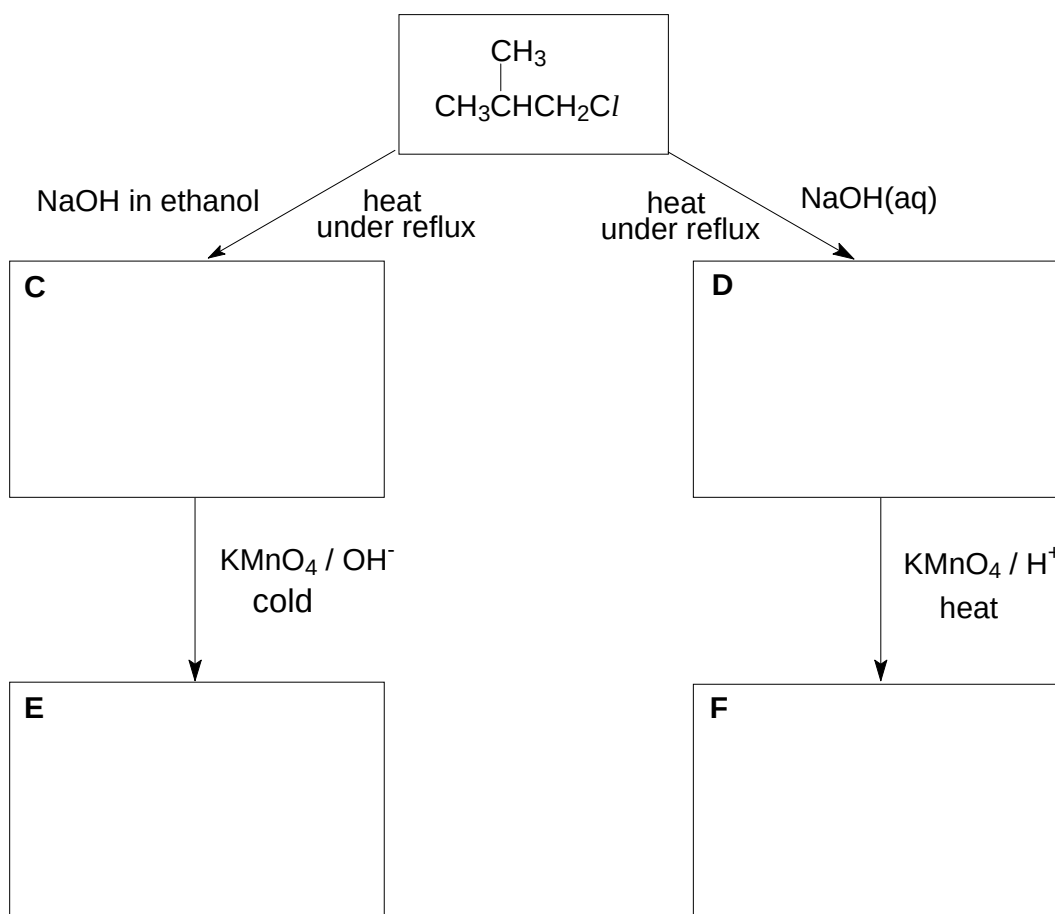
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[3]

- (b) Chlorine reacts with methylpropane in the presence of UV light to form a mixture of products. One of the products, 1-chloro-2-methylpropane, undergoes a sequence of reactions as shown below.



In the appropriate boxes, draw the structures of compounds **C**, **D**, **E** and **F**.

[4]

- (c) 1-chloro-2-methylpropane and 2-chlorobutane are isomers.

- (i) State the type of isomerism exhibited by 1-chloro-2-methylpropane and 2-chlorobutane.

.....

- (ii) Why is 2-chlorobutane considered an environmental hazard?

.....

[2]

[Total: 9]

- 4 Bromoalkanes **G** and **H** are structural isomers with the molecular formula $\text{C}_5\text{H}_{11}\text{Br}$. Two experiments were carried out to investigate the kinetics of the reaction of **G** and **H** with aqueous sodium hydroxide separately.

- (a) The following initial rates were obtained using different concentrations of **G** and aqueous sodium hydroxide.

experiment	[G] / mol dm^{-3}	[NaOH] / mol dm^{-3}	initial rate / $\text{mol dm}^{-3} \text{ min}^{-1}$
1	0.10	0.10	5.56×10^{-4}
2	0.20	0.05	5.56×10^{-4}
3	0.30	0.10	1.67×10^{-3}

- (i) Use the data to determine the order of reaction with respect to

I bromoalkane **G**

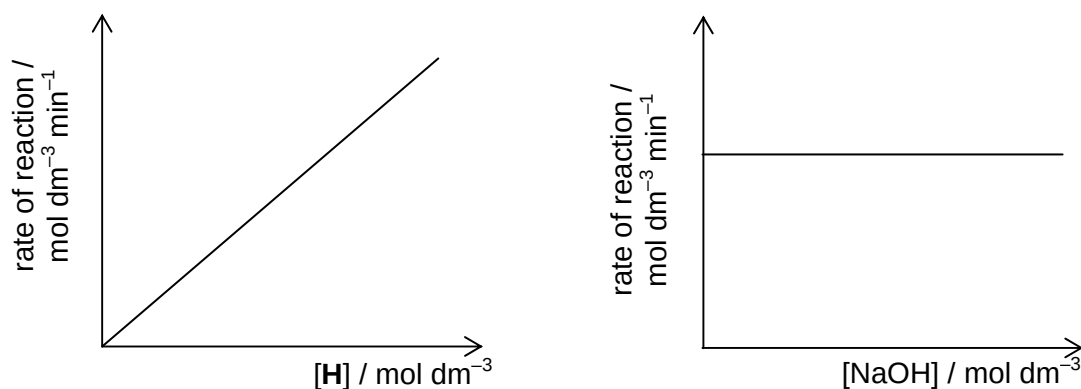
II aqueous sodium hydroxide

- (ii) Use these answers to write a rate equation for the reaction.

- (iii) Hence calculate the rate constant for the reaction, stating its units.

[5]

- (b) The following graphs were obtained in the investigation of the kinetics of the reaction of **H** and aqueous sodium hydroxide.



- (i) Using the graphs above, deduce the rate equation for the reaction of **H** with aqueous sodium hydroxide.

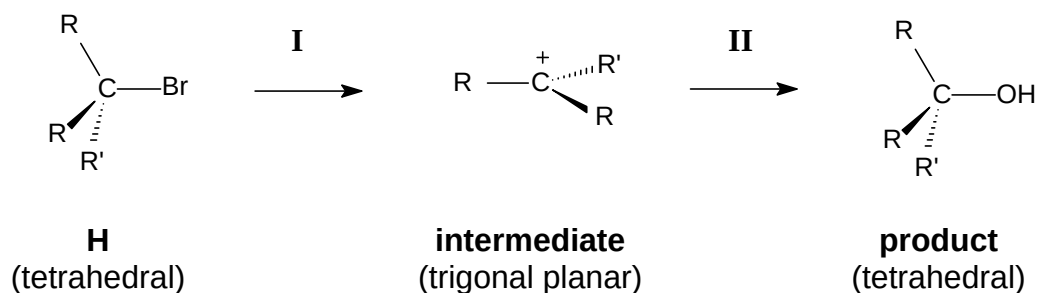
- (ii) Hence, explain whether the reaction of aqueous sodium hydroxide with bromoalkanes **G** or **H** proceeds by the same mechanism.

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.....

[2]

- (c) When **H** reacts with aqueous sodium hydroxide, there is a change in structure from reactant to product, as shown below.



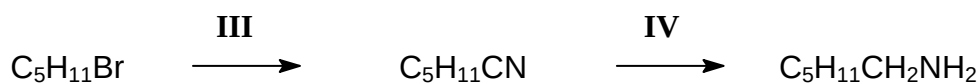
- (i) Identify the change in the hybridisation of the reactive carbon atom in steps **I** and **II**.

Step	Change in hybridisation of carbon
I	
II	

- (ii) Hence sketch the shape of one hybrid orbital of the reactive carbon atom in reactant **H**.

[2]

- (d) Bromoalkane is a useful intermediate for synthesising other organic compounds. The reaction scheme below shows one such example.



- (i) Suggest reagents and conditions for steps **III** and **IV**.

III :

IV :

- (ii) State the type of reaction occurring in step **IV**.

[3]

[Total: 12]

Section B

Answer **two** of the three questions in this section on separate writing paper.

- 5 Compounds of sulfur have many uses in everyday life which include disinfectants, matches and in making of paper.

(a) Phosphorus sulfide, P_4S_3 , is used in small amounts in the tip of a match stick. On striking a match stick, one mole of this compound burns to form sulfur dioxide and a solid oxide in the molar ratio of 3:1.

(i) Construct an equation for this reaction.

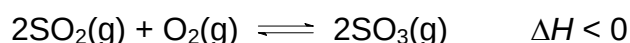
(ii) Both oxides formed in **(a)(i)** dissolve in water to give acidic solutions. Construct an equation for the reaction of each oxide with water and give an approximate pH for each of the solution formed.

(iii) The melting points of the two oxides formed in **(a)(i)** differ significantly from that of silicon (IV) oxide (SiO_2).

Briefly relate this observation to their structure and bonding.

[6]

(b) The key reaction during the Contact process for manufacturing sulfuric acid is as follows.



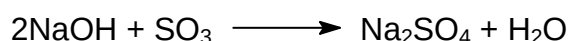
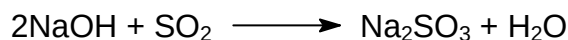
When 34.6 mol of SO_2 and 17.3 mol of O_2 are passed over a catalyst in a 1 dm^3 vessel at 450°C , the amount of SO_3 at equilibrium is found to be 32.8 mol.

(i) Calculate the value and state the units of the equilibrium constant, K_c , for this reaction.

(ii) Calculate the percentage of SO_2 in an SO_2 – SO_3 mixture at 450°C when $[O_2] = 10 \text{ mol dm}^{-3}$.

- (iii) The total SO_2 and SO_3 content in the mixture in (b)(ii) can be found by dissolving the mixture in water and titrating against aqueous sodium hydroxide of known concentration.

When a 100 cm^3 sample of the mixture was titrated against $0.200 \text{ mol dm}^{-3}$ sodium hydroxide, 35.00 cm^3 of aqueous sodium hydroxide was required to neutralise all the oxides of sulfur according to the following equations:

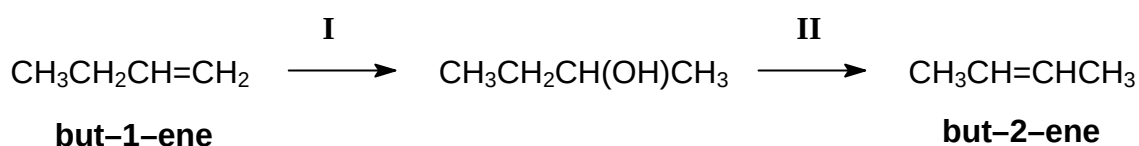


Calculate the number of moles of sodium hydroxide required and the total number of moles of SO_2 and SO_3 present in the sample. Hence, by using the answer in (b)(ii), calculate the concentration of SO_3 , in mol dm^{-3} , in the sample.

- (iv) State how the position of equilibrium and rate of achieving equilibrium would be affected by a decrease in temperature.

[10]

- (c) Concentrated sulfuric acid can be used under two different sets of reaction conditions to synthesise but-2-ene from but-1-ene via a two-step reaction scheme as shown below.



- (i) Describe the condition necessary for step II.
- (ii) But-2-ene occurs in two isomeric forms but cyclobutene occurs in only one form. Draw the displayed formulae of the two isomers of but-2-ene and explain why such isomerism is not possible with cyclobutene.

[4]

[Total: 20]

6 (a) Sodium hydroxide is a common reagent used in laboratory. It can be prepared by dissolving sodium oxide in water.

- (i) With the aid of an appropriate equation, explain why sodium hydroxide can be used to separate a mixture of magnesium oxide and aluminium oxide.
- (ii) The lattice energy of sodium oxide is -742 kJ mol^{-1} . Explain how you would expect the magnitude of the lattice energy of magnesium oxide to compare with that of sodium oxide.

[4]

(b) Compound **J** ($\text{C}_8\text{H}_{16}\text{O}_2$) is a neutral extract. A chemical analysis of **J** is carried out, using sodium hydroxide as one of the reagents. The following results are obtained.

- **J** is insoluble in both water and dilute aqueous sodium hydroxide.
- When heated under reflux with dilute aqueous sodium hydroxide followed by acidification, **J** gives two compounds, **K** ($\text{C}_5\text{H}_{12}\text{O}$) and **L** ($\text{C}_3\text{H}_6\text{O}_2$).

Reaction of **K** with aqueous iodine in sodium hydroxide produces a yellow precipitate. Heating **K** over Al_2O_3 produces compound **M** (C_5H_{10}). Strong heating of **M** with acidified potassium manganate(VII) produces propanone as one of the products.

Effervescence is observed when aqueous sodium carbonate is added to **L**.

Use all of the above information to identify the organic compounds **J**, **K**, **L** and **M**, and draw a structural formula for each of them.

Write equations where appropriate to show the reactions that are occurring. Clearly show all of the deductions that you make from the information that you have been given. Full marks cannot be gained by only giving the structures required.

[13]

- (c) (i) With reference to compound **M** (C_5H_{10}), explain what is meant by the term *standard enthalpy change of combustion*?
- (ii) Assuming that air contains one-fifth oxygen by volume and that both gas volumes are measured at the same temperature and pressure, calculate the volume of air required for complete combustion of 10 cm^3 of **M**.

[3]

[Total:20]

- 7 (a) (i) What do you understand by the terms *strong* and *weak* when applied to acids?
- (ii) It is found that an aqueous solution of benzoic acid of concentration 0.10 mol dm^{-3} has a pH of 2.6. Explain, with reasons, whether benzoic acid is a strong or a weak acid.

[2]

- (b) A sample of 20.0 cm^3 aqueous solution of 0.1 mol dm^{-3} benzoic acid was titrated with 0.15 mol dm^{-3} potassium hydroxide solution.

- (i) Calculate the pH of the 0.15 mol dm^{-3} potassium hydroxide solution.
- (ii) Calculate the volume of potassium hydroxide required for neutralisation with the benzoic acid.
- (iii) Suggest **two** reasons why phenolphthalein is a suitable indicator for this titration.

[4]

- (c) A buffer containing potassium benzoate and benzoic acid was formed during the titration in (b). With the aid of **two** equations, explain how a solution of potassium benzoate and benzoic acid can control pH.

[2]

- (d) The enthalpy changes of neutralisation of potassium hydroxide by two acids are as follows:

hydrochloric acid	$-57.3 \text{ kJ mol}^{-1}$
nitric acid	$-57.3 \text{ kJ mol}^{-1}$

Explain how you would expect the enthalpy change of neutralisation of potassium hydroxide by benzoic acid to compare with these values.

[2]

- (e) The relative molecular mass of benzoic acid is found to be approximately 244 at 25°C when it is dissolved in benzene.

- (i) Draw a diagram to explain this phenomenon.
- (ii) State and explain what will be observed when a piece of dry blue litmus paper is dropped into this solution.

[2]

- (f) Benzoic acid crystals can be synthesized from methylbenzene in the laboratory.

- (i) Suggest the reagents and conditions required for this synthesis.
- (ii) Write a balanced equation for this reaction.

[2]

- (g) When methylbenzene is treated with chlorine under two different conditions, three isomeric monochlorides, **P**, **Q** and **R** are formed.

- (i) Deduce the structures of isomers **P**, **Q** and **R**.
- (ii) Describe the two conditions necessary to form the monochlorides.
- (iii) Suggest how, if any, methylbenzene and any **one** of the isomers in (g)(i) might differ in their boiling points? Explain your answers.

[6]

[Total:20]

End of Paper